

# Climate 5000 Series - Gen 4

## 18K BTU Single Zone Max Performance System

### Wall Mounted IDU

Engineering  
Submittal  
Sheet



# BOSCH

#### Overview and Project Information



|                    |  |
|--------------------|--|
| Project name       |  |
| Location           |  |
| Architect          |  |
| Engineer           |  |
| General contractor |  |
| Reference #        |  |

#### 18K Btu Single Zone Max Performance System - Wall Mounted IDU

|                                          |                      |                          |
|------------------------------------------|----------------------|--------------------------|
| Indoor part number                       |                      | 8-733-962-674            |
| Indoor model number                      |                      | BMS500-AAU018-1AHWXD     |
| Outdoor part number                      |                      | 8-733-962-690            |
| Outdoor model number                     |                      | BMS500-AAS018-1CSXHD     |
| <b>Efficiency Data</b>                   |                      | <b>AHRI 210/240 2023</b> |
| SEER2                                    |                      | 21.4                     |
| EER2                                     |                      | 12.5                     |
| HSPF2 Region IV                          |                      | 11.0                     |
| HSPF2 Region V                           |                      | 8.5                      |
| COP(47F)                                 |                      | 3.26                     |
| COP (5F)                                 |                      | 1.92                     |
| <b>Performance Data</b>                  |                      | <b>AHRI 210/240 2023</b> |
| Cooling                                  | Capacity Range       | 8800~19400               |
|                                          | Rated Capacity       | 18000                    |
|                                          | Current              | 6.4                      |
|                                          | Input Power          | 1440                     |
| Heating                                  | Capacity Range       | 11200~19500              |
|                                          | Rated Capacity (47F) | 18000                    |
|                                          | Rated Capacity (5F)  | 18000                    |
|                                          | Current              | 7.2                      |
|                                          | Input Power          | 1618                     |
| <b>Electrical Data</b>                   |                      |                          |
| Power supply (V / Ø / Hz)                |                      | 208-230 / 1 / 60         |
| Min. circuit ampacity indoor/outdoor (A) |                      | 3.0 / 19.0               |
| Max. fuse indoor/outdoor (A)             |                      | 15.0 / 20.0              |
| <b>Compressor Data</b>                   |                      |                          |
| Type:                                    |                      | Rotary                   |
| Refrigerant oil /oil charge (ml)         |                      | VG74 / 620               |
| Input (W)                                |                      | 805 / 2045               |
| Rated current (RLA/A)                    |                      | 4.15 / 9.30              |
| <b>Operation Range</b>                   |                      |                          |
| ODU                                      | Cooling (°F / °C)    | -22~122 / -30~50         |
|                                          | Heating (°F / °C)    | -22~75 / -30~24          |
| IDU                                      | Cooling (°F / °C)    | 60~90 / 16~32            |
|                                          | Heating (°F / °C)    | 32~86 / 0~30             |

#### Piping

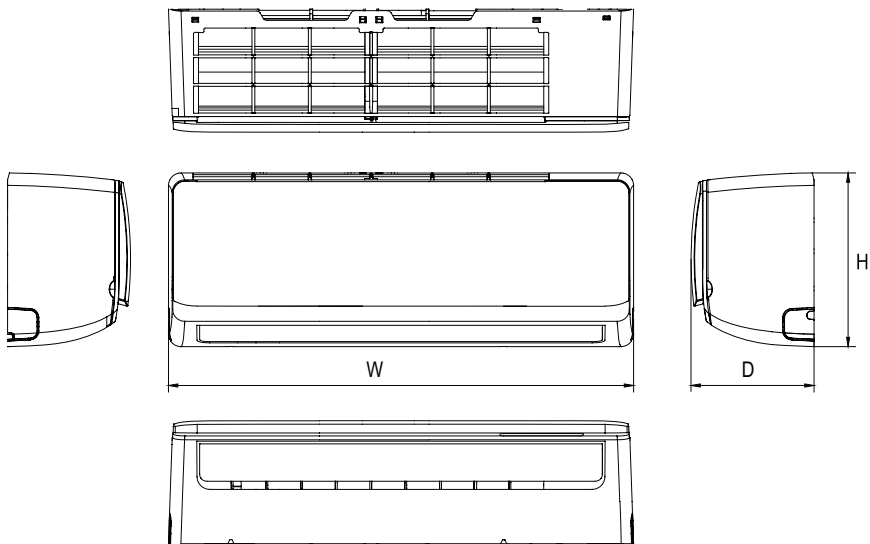
|                                                |                                             |                                   |
|------------------------------------------------|---------------------------------------------|-----------------------------------|
| Refrigerant precharge (oz/g)                   |                                             | 55.38 / 1570.00                   |
| Length for precharge (ft/m)                    |                                             | 24.6 / 7.5                        |
| Liquid side (in/mm)                            |                                             | 1/4" / Ø6.35                      |
| Gas side (in/mm)                               |                                             | 1/2" / Ø12.7                      |
| Additional Charge of Refrigerant (oz/ft)/(g/m) |                                             | 0.16 / 15                         |
| Max. refrigerant pipe length (ft/m)            |                                             | 98.40 / 30                        |
| Max. difference in height (ft/m)               |                                             | 65.6 / 20                         |
| <b>System Data</b>                             |                                             |                                   |
| Refrigerant type                               |                                             | R454B                             |
| Design pressure (PSIG)                         |                                             | 550 / 340                         |
| Dehumidification (pts/hr)                      |                                             | 4.64                              |
| IDU                                            | Fan motor model                             | ZKFP-30-8-3-10                    |
|                                                | Fan motor qty (ea)                          | 1                                 |
|                                                | Fan motor input (W)                         | 36.0                              |
|                                                | Fan speed (RPM)                             | 1190 / 994 / 896                  |
|                                                | Air flow, Turbo / H / M / L / Si (CFM)      | 635.7 / 500.3 / 359.0 / 294.3 / / |
|                                                | Approx. Air Throw Range ft (m)              | 26.2 / 8                          |
|                                                | Noise level, Turbo / H / M / L / Si (dB(A)) | 49.5 / 45.0 / 38.0 / 31.5 / 19.5  |
|                                                | Net weight (lbs / kg)                       | 27.34 / 12.4                      |
| ODU                                            | Gross weight (lbs / kg)                     | 36.60 / 16.6                      |
|                                                | Fan motor model                             | ZKFN-80-8-3                       |
|                                                | Fan motor qty (ea)                          | 1                                 |
|                                                | Fan motor input (W)                         | 115                               |
|                                                | Fan motor speed, H / M / L (RPM)            | 780 / 700 / 550                   |
|                                                | Air flow, Max (CFM)                         | 1764.7                            |
|                                                | Noise level, Max (dB(A))                    | 59                                |
|                                                | Net weight (lbs / kg)                       | 99.87 / 45.3                      |
| Accessories                                    | Gross weight (lbs / kg)                     | 107.8 / 48.9                      |
|                                                | BMS-WT2-XXD (8733962707)                    | Wired Wall Thermostat             |
| Standard Parts                                 | BMS-DRY-XXD (8733969263)                    | 24V Interface                     |
|                                                | BMS-PK2-7X1 (8733951011)                    | 25ft Lineset                      |
| 8-733-969-088                                  |                                             | Remote Control                    |

Climate 5000 Series - Gen 4

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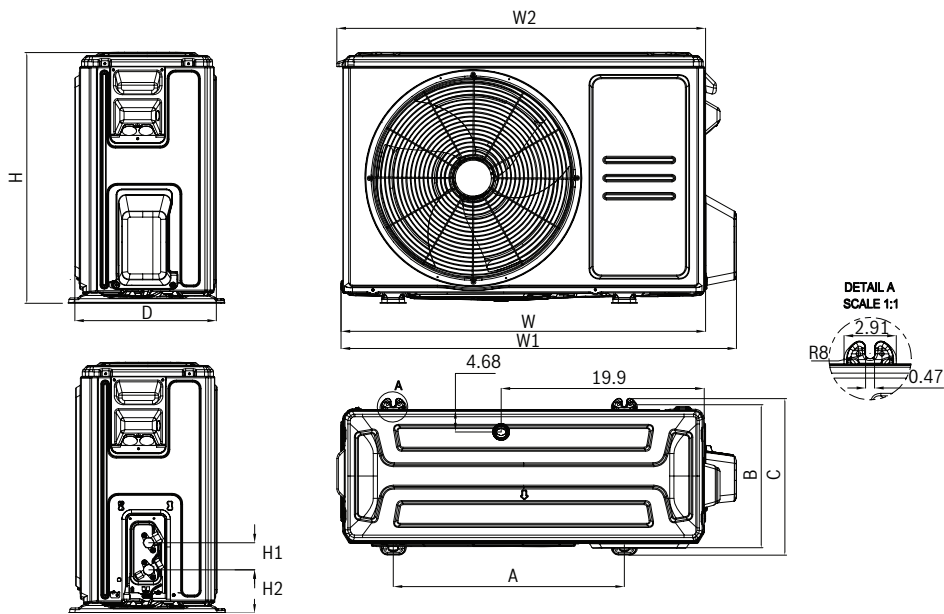


Indoor Unit Dimensions and Connections



| Model                | W (in) | D (in) | H (in) |
|----------------------|--------|--------|--------|
| BMS500-AAU018-1AHWXD | 38.0   | 9.8    | 12.56  |

Outdoor Unit Dimensions and Connections



| Model                | W (in) | D (in) | H (in) | W1 (in) | W2 (in) | H1 (in) | H2 (in) | A (in) | B (in) | C (in) |
|----------------------|--------|--------|--------|---------|---------|---------|---------|--------|--------|--------|
| BMS500-AAS018-1CSXHD | 35.04  | 13.46  | 26.5   | 37.6    | 35.24   | 2.36    | 4.25    | 26.1   | 13.7   | 14.96  |

## Climate 5000 Series - Gen 4

## 18K Btu Single Zone Max Performance System - Wall Mounted IDU


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| Cooling Performance - Indoor Dry Bulb Temperature at 635 CM |             |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |
|-------------------------------------------------------------|-------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| ODU DB(°F)                                                  | IDU WB (°F) | 60.8°F(16.0°C)  |                 |                 |                 |                 | 64.4°F(18.0°C)  |                 |                 |                 |                 | 67°F(19.4°C)    |                 |                 |                 |                 | 71.6°F(22.0°C)  |                 |                 |                 |                 |
|                                                             | IDU DB (°F) | 73.4°F (23.0°C) | 77.0°F (25.0°C) | 80.0°F (26.7°C) | 80.6°F (27.0°C) | 84.2°F (29.0°C) | 73.4°F (23.0°C) | 77.0°F (25.0°C) | 80.0°F (26.7°C) | 80.6°F (27.0°C) | 84.2°F (29.0°C) | 73.4°F (23.0°C) | 77.0°F (25.0°C) | 80.0°F (26.7°C) | 80.6°F (27.0°C) | 84.2°F (29.0°C) | 73.4°F (23.0°C) | 77.0°F (25.0°C) | 80.0°F (26.7°C) | 80.6°F (27.0°C) | 84.2°F (29.0°C) |
| -22°F (-30.0°C)                                             | TC          | 14208           | 14202           | 14404           | 14464           | 14862           | 14824           | 14812           | 14803           | 14801           | 14789           | 15268           | 15248           | 15234           | 15232           | 15217           | 16104           | 16100           | 16067           | 16061           | 16027           |
|                                                             | S/T         | 0.80            | 0.91            | 0.99            | 0.99            | 0.99            | 0.64            | 0.75            | 0.84            | 0.86            | 0.97            | 0.55            | 0.65            | 0.74            | 0.75            | 0.86            | 0.39            | 0.47            | 0.55            | 0.57            | 0.67            |
|                                                             | PI          | 0.69            | 0.69            | 0.69            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.68            | 0.69            | 0.69            | 0.69            | 0.69            | 0.69            |
| -13°F (-25.0°C)                                             | TC          | 15392           | 15385           | 15605           | 15669           | 16101           | 16060           | 16046           | 16036           | 16035           | 16022           | 16540           | 16519           | 16503           | 16501           | 16485           | 17446           | 17442           | 17406           | 17399           | 17362           |
|                                                             | S/T         | 0.80            | 0.91            | 1.00            | 1.00            | 1.00            | 0.65            | 0.75            | 0.85            | 0.86            | 0.97            | 0.55            | 0.65            | 0.74            | 0.76            | 0.86            | 0.39            | 0.47            | 0.56            | 0.57            | 0.67            |
|                                                             | PI          | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            | 0.71            |
| -4°F (-20.0°C)                                              | TC          | 16576           | 16569           | 16805           | 16875           | 17339           | 17295           | 17281           | 17270           | 17268           | 17254           | 17812           | 17790           | 17773           | 17770           | 17753           | 18788           | 18783           | 18745           | 18738           | 18698           |
|                                                             | S/T         | 0.80            | 0.92            | 1.00            | 1.00            | 1.00            | 0.65            | 0.76            | 0.85            | 0.87            | 0.98            | 0.55            | 0.65            | 0.74            | 0.76            | 0.86            | 0.39            | 0.48            | 0.56            | 0.57            | 0.67            |
|                                                             | PI          | 0.74            | 0.74            | 0.74            | 0.74            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            |
| 0°F (-17.8°C)                                               | TC          | 17126           | 17116           | 17434           | 17507           | 17988           | 17861           | 17841           | 17827           | 17825           | 18005           | 18386           | 18360           | 18337           | 18334           | 18326           | 19375           | 19370           | 19369           | 19371           | 19384           |
|                                                             | S/T         | 0.81            | 0.93            | 1.00            | 1.00            | 1.00            | 0.66            | 0.77            | 0.86            | 0.88            | 1.00            | 0.55            | 0.66            | 0.75            | 0.77            | 0.87            | 0.39            | 0.48            | 0.57            | 0.58            | 0.68            |
|                                                             | PI          | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.75            | 0.75            | 0.75            | 0.75            | 0.75            |
| 5°F (-15.0°C)                                               | TC          | 17618           | 17678           | 18082           | 18166           | 18725           | 18444           | 18448           | 18450           | 18469           | 18744           | 19034           | 19038           | <b>19042</b>    | 19043           | 19068           | 20150           | 20144           | 20154           | 20157           | 20165           |
|                                                             | S/T         | 0.82            | 0.94            | 1.00            | 1.00            | 1.00            | 0.66            | 0.78            | 0.87            | 0.88            | 1.00            | 0.56            | 0.67            | 0.76            | 0.78            | 0.88            | 0.39            | 0.48            | 0.57            | 0.59            | 0.69            |
|                                                             | PI          | 0.75            | 0.75            | 0.75            | 0.75            | 0.76            | 0.75            | 0.75            | 0.75            | 0.75            | 0.76            | 0.76            | 0.76            | <b>0.76</b>     | 0.76            | 0.76            | 0.77            | 0.77            | 0.77            | 0.77            | 0.77            |
| 14°F (-10.0°C)                                              | TC          | 18304           | 18294           | 18668           | 18755           | 19337           | 19212           | 19195           | 19182           | 19180           | 19360           | 19871           | 19846           | 19825           | 19822           | 19802           | 21129           | 21122           | 21082           | 21074           | 21029           |
|                                                             | S/T         | 0.81            | 0.93            | 1.00            | 1.00            | 1.00            | 0.65            | 0.76            | 0.86            | 0.87            | 1.00            | 0.55            | 0.66            | 0.75            | 0.76            | 0.87            | 0.39            | 0.48            | 0.56            | 0.57            | 0.67            |
|                                                             | PI          | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.73            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            |
| 17°F (-8.3°C)                                               | TC          | 18488           | 18549           | 18977           | 19070           | 19689           | 19409           | 19417           | 19420           | 19436           | 19710           | 20071           | 20078           | <b>20086</b>    | 20087           | 20112           | 21346           | 21340           | 21337           | 21341           | 21359           |
|                                                             | S/T         | 0.82            | 0.93            | 1.00            | 1.00            | 1.00            | 0.66            | 0.77            | 0.87            | 0.88            | 1.00            | 0.55            | 0.66            | 0.76            | 0.77            | 0.87            | 0.39            | 0.48            | 0.57            | 0.58            | 0.68            |
|                                                             | PI          | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | <b>0.73</b>     | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            |
| 23°F (-5.0°C)                                               | TC          | 18722           | 18802           | 19289           | 19384           | 20023           | 19670           | 19676           | 19678           | 19704           | 20044           | 20348           | 20355           | 20361           | 20361           | 20394           | 21636           | 21630           | 21645           | 21648           | 21660           |
|                                                             | S/T         | 0.83            | 0.95            | 1.00            | 1.00            | 1.00            | 0.66            | 0.78            | 0.88            | 0.89            | 1.00            | 0.56            | 0.67            | 0.76            | 0.78            | 0.89            | 0.39            | 0.48            | 0.57            | 0.59            | 0.69            |
|                                                             | PI          | 0.73            | 0.73            | 0.73            | 0.73            | 0.74            | 0.73            | 0.73            | 0.73            | 0.73            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.74            | 0.75            | 0.75            | 0.75            | 0.75            | 0.75            |
| 35°F (1.7°C)                                                | TC          | 19369           | 19370           | 19484           | 19538           | 20168           | 20274           | 20274           | 20275           | 20275           | 20329           | 20909           | 20908           | <b>20909</b>    | 20909           | 20911           | 22086           | 22081           | 22080           | 22079           | 22081           |
|                                                             | S/T         | 0.78            | 0.88            | 0.97            | 0.99            | 1.00            | 0.63            | 0.73            | 0.82            | 0.83            | 0.93            | 0.54            | 0.63            | 0.72            | 0.73            | 0.83            | 0.39            | 0.46            | 0.54            | 0.56            | 0.65            |
|                                                             | PI          | 0.69            | 0.69            | 0.69            | 0.70            | 0.72            | 0.73            | 0.73            | 0.73            | 0.73            | 0.73            | 0.76            | 0.76            | <b>0.76</b>     | 0.76            | 0.76            | 0.83            | 0.83            | 0.83            | 0.83            | 0.83            |
| 41°F (5.0°C)                                                | TC          | 19121           | 19121           | 19273           | 19407           | 19981           | 20010           | 20010           | 20011           | 20011           | 20081           | 20634           | 20633           | 20634           | 20634           | 20636           | 21792           | 21787           | 21785           | 21784           | 21787           |
|                                                             | S/T         | 0.78            | 0.89            | 0.98            | 1.00            | 1.00            | 0.64            | 0.74            | 0.83            | 0.84            | 0.94            | 0.54            | 0.64            | 0.72            | 0.74            | 0.84            | 0.39            | 0.47            | 0.55            | 0.56            | 0.66            |
|                                                             | PI          | 0.74            | 0.74            | 0.75            | 0.75            | 0.78            | 0.78            | 0.78            | 0.78            | 0.78            | 0.79            | 0.82            | 0.82            | 0.82            | 0.82            | 0.82            | 0.89            | 0.89            | 0.89            | 0.89            | 0.89            |
| 47°F (8.3°C)                                                | TC          | 18872           | 18884           | 19137           | 19223           | 19792           | 19747           | 19746           | 19746           | 19746           | 19838           | 20360           | 20358           | <b>20359</b>    | 20359           | 20359           | 21500           | 21494           | 21492           | 21492           | 21494           |
|                                                             | S/T         | 0.79            | 0.90            | 1.00            | 1.00            | 1.00            | 0.64            | 0.75            | 0.84            | 0.85            | 0.96            | 0.54            | 0.65            | 0.73            | 0.75            | 0.85            | 0.39            | 0.47            | 0.55            | 0.57            | 0.67            |
|                                                             | PI          | 0.79            | 0.79            | 0.80            | 0.81            | 0.84            | 0.84            | 0.84            | 0.84            | 0.84            | 0.85            | 0.88            | 0.88            | <b>0.88</b>     | 0.88            | 0.88            | 0.96            | 0.96            | 0.96            | 0.96            | 0.96            |
| 50°F (10.0°C)                                               | TC          | 18525           | 19035           | 19677           | 19791           | 20559           | 19571           | 19582           | 19717           | 19779           | 20578           | 20334           | 20361           | 20371           | 20373           | 20568           | 21736           | 21793           | 21849           | 21856           | 21883           |
|                                                             | S/T         | 0.90            | 1.00            | 1.00            | 1.00            | 1.00            | 0.71            | 0.85            | 0.96            | 0.99            | 1.00            | 0.59            | 0.72            | 0.83            | 0.85            | 0.99            | 0.40            | 0.50            | 0.60            | 0.62            | 0.74            |
|                                                             | PI          | 0.91            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            | 0.90            |
| 59°F (15.0°C)                                               | TC          | 18498           | 19147           | 19827           | 19948           | 20759           | 19572           | 19599           | 19821           | 19969           | 20780           | 20380           | 20400           | 20407           | 20437           | 20797           | 21884           | 21927           | 21971           | 21976           | 21994           |
|                                                             | S/T         | 0.92            | 1.00            | 1.00            | 1.00            | 1.00            | 0.73            | 0.86            | 1.00            | 1.00            | 1.00            | 0.60            | 0.73            | 0.85            | 0.86            | 1.00            | 0.40            | 0.51            | 0.61            | 0.63            | 0.76            |
|                                                             | PI          | 0.97            | 0.97            | 0.97            | 0.97            | 0.98            | 0.97            | 0.97            | 0.97            | 0.97            | 0.98            | 0.98            | 0.98            | 0.98            | 0.98            | 0.98            | 0.98            | 0.98            | 0.98            | 0.98            | 0.98            |
| 68°F (20.0°C)                                               | TC          | 18463           | 19251           | 19965           | 20092           | 20945           | 19546           | 19590           | 19987           | 20114           | 20966           | 20391           | 20407           | 20443           | 20461           | 20983           | 21972           | 22012           | 22047           | 22051           | 22062           |
|                                                             | S/T         | 0.95            | 1.00            | 1.00            | 1.00            | 1.00            | 0.74            | 0.89            | 1.00            | 1.00            | 1.00            | 0.61            | 0.75            | 0.87            | 0.89            | 1.00            | 0.40            | 0.51            | 0.63            | 0.65            | 0.78            |
|                                                             | PI          | 1.03            | 1.04            | 1.04            | 1.05            | 1.05            | 1.04            | 1.04            | 1.04            | 1.05            | 1.05            | 1.05            | 1.05            | 1.05            | 1.05            | 1.05            | 1.06            | 1.06            | 1.06            | 1.06            | 1.06            |
| 77°F (25.0°C)                                               | TC          | 18413           | 19299           | 20055           | 20188           | 21091           | 19520           | 19582           | 20077           | 20211           | 21112           | 20404           | 20427           | 20479           | 20502           | 21129           | 22037           | 22104           | 22155           | 22161           | 22181           |
|                                                             | S/T         | 0.97            | 1.00            | 1.00            | 1.00            | 1.00            | 0.75            | 0.90            | 1.00            | 1.00            | 1.00            | 0.61            | 0.76            | 0.88            | 0.90            | 1.00            | 0.40            | 0.52            | 0.63            | 0.65            | 0.79            |
|                                                             | PI          | 1.11            | 1.11            | 1.12            | 1.12            | 1.12            | 1.11            | 1.11            | 1.12            | 1.12            | 1.12            | 1.12            | 1.12            | 1.12            | 1.12            | 1.12            | 1.13            | 1.13            | 1.13            | 1.13            | 1.13            |
| 82°F (27.8°C)                                               | TC          | 18255           | 19187           | 19961           | 20096           | 21016           | 19345           | 19419           | 19978           | 20119           | 21038           | 20244           | 20270           | 20333           | 20358           | 21057           | 21888           | 21971           | 22028           | 22034           | 22058           |
|                                                             | S/T         | 0.99            | 1.00            | 1.00            | 1.00            | 1.00            | 0.76            | 0.91            | 1.00            | 1.00            | 1.00            | 0.62            | 0.77            | 0.89            | 0.92            | 1.00            | 0.40            | 0.52            | 0.64            | 0.66            | 0.80            |
|                                                             | PI          | 1.20            | 1.21            | 1.21            | 1.21            | 1.22            | 1.21            | 1.21            | 1.21            | 1.21            | 1.22            | 1.21            | 1.21            | 1.21            | 1.21            | 1.22            | 1.23            | 1.23            | 1.23            | 1.23            | 1.22            |
| 86°F (30.0°C)                                               | TC          | 18025           | 18925           | 19699           | 19837           | 20761           | 19009           | 19097           | 19721           | 19859           | 20782           | 19910           | 19933           | 20006           | 20035           | 20800           | 21555           | 21638           | 21690           | 21693           | 21715           |
|                                                             | S/T         | 1.00            | 1.00            | 1.00            | 1.00            | 1.00            | 0.77            | 0.93            | 1.00            | 1.00            | 1.00            | 0.62            | 0.78            | 0.91            | 0.93            | 1.00            | 0.40            | 0.52            | 0.64            | 0.66            | 0.81            |
|                                                             | PI          | 1.34            | 1.35            | 1.35            | 1.35            | 1.36            | 1.35            | 1.35            | 1.35            | 1.35            | 1.36            | 1.36            | 1.36            | 1.36            | 1.36            | 1.36            | 1.37            | 1.37            | 1.37            | 1.37            | 1.37            |
| 95°F (35.0°C)                                               | TC          | 17372           | 17466           | 180308          |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |

# Climate 5000 Series - Gen 4

## 18K Btu Single Zone Max Performance System - Wall Mounted IDU



# BOSCH

| Heating Performance - Indoor Dry Bulb Temperature at 635 CM |                                   |                  |                  |                    |                    |                          |                  |                  |                    |                    |
|-------------------------------------------------------------|-----------------------------------|------------------|------------------|--------------------|--------------------|--------------------------|------------------|------------------|--------------------|--------------------|
| Outdoor DB(°F)                                              | TC Total Heating Capacity(BTU/hr) |                  |                  |                    |                    | PI Power Input(kW)       |                  |                  |                    |                    |
|                                                             | Indoor Conditions(DB °F)          |                  |                  |                    |                    | Indoor Conditions(DB °F) |                  |                  |                    |                    |
|                                                             | 60.8°F<br>(16.0°C)                | 68°F<br>(20.0°C) | 70°F<br>(21.1°C) | 71.6°F<br>(22.0°C) | 75.2°F<br>(24.0°C) | 60.8°F<br>(16.0°C)       | 68°F<br>(20.0°C) | 70°F<br>(21.1°C) | 71.6°F<br>(22.0°C) | 75.2°F<br>(24.0°C) |
| <b>-22°F(-30.0°C)</b>                                       | 9595                              | 9403             | <b>9356</b>      | 9318               | 9234               | 1.91                     | 1.93             | <b>1.93</b>      | 1.93               | 1.94               |
| <b>-13°F(-25.0°C)</b>                                       | 11579                             | 11347            | <b>11290</b>     | 11245              | 11144              | 2.01                     | 2.03             | <b>2.03</b>      | 2.03               | 2.04               |
| <b>-4°F(-20.0°C)</b>                                        | 14763                             | 14468            | <b>14395</b>     | 14338              | 14209              | 2.18                     | 2.19             | <b>2.20</b>      | 2.20               | 2.21               |
| 0°F(-17.8°C)                                                | 16480                             | 16150            | 16069            | 16005              | 15861              | 2.17                     | 2.18             | 2.19             | 2.19               | 2.20               |
| <b>5°F(-15.0°C)</b>                                         | 18626                             | 18253            | <b>18162</b>     | 18089              | 17927              | 2.16                     | 2.17             | <b>2.18</b>      | 2.18               | 2.19               |
| 14°F(-10.0°C)                                               | 18513                             | 18143            | 18052            | 17980              | 17818              | 2.41                     | 2.43             | 2.43             | 2.43               | 2.44               |
| <b>17°F(-8.3°C)</b>                                         | 18475                             | 18106            | <b>18015</b>     | 17943              | 17782              | 2.49                     | 2.51             | <b>2.51</b>      | 2.52               | 2.53               |
| 22°F(-5.6°C)                                                | 18406                             | 18038            | 17948            | 17876              | 17715              | 2.36                     | 2.38             | 2.39             | 2.39               | 2.40               |
| 27°F(-2.8°C)                                                | 18337                             | 17971            | 17881            | 17809              | 17649              | 2.23                     | 2.25             | 2.26             | 2.26               | 2.27               |
| 32°F(0.0°C)                                                 | 18268                             | 17903            | 17813            | 17742              | 17583              | 2.11                     | 2.12             | 2.13             | 2.13               | 2.14               |
| <b>35°F(1.7°C)</b>                                          | 18227                             | 17862            | <b>17773</b>     | 17702              | 17543              | 2.03                     | 2.05             | <b>2.05</b>      | 2.06               | 2.06               |
| 37°F(2.8°C)                                                 | 18652                             | 18278            | 18187            | 18114              | 17951              | 2.00                     | 2.02             | 2.02             | 2.03               | 2.03               |
| 42°F(5.6°C)                                                 | 19713                             | 19319            | 19222            | 19145              | 18973              | 1.93                     | 1.94             | 1.95             | 1.95               | 1.96               |
| 44.6°F(7.0°C)                                               | 20137                             | 19735            | 19636            | 19558              | 19381              | 1.90                     | 1.91             | 1.92             | 1.92               | 1.93               |
| <b>47°F(8.3°C)</b>                                          | 20774                             | 20359            | <b>20257</b>     | 20176              | 19994              | 1.85                     | 1.87             | <b>1.87</b>      | 1.88               | 1.88               |
| 52°F(11.1°C)                                                | 20045                             | 19644            | 19546            | 19468              | 19293              | 1.63                     | 1.65             | 1.65             | 1.65               | 1.66               |
| 57°F(13.9°C)                                                | 19316                             | 18930            | 18835            | 18760              | 18591              | 1.42                     | 1.43             | 1.43             | 1.43               | 1.44               |
| 59°F(15.0°C)                                                | 19024                             | 18644            | 18550            | 18476              | 18310              | 1.33                     | 1.34             | 1.34             | 1.34               | 1.35               |
| 62°F(16.7°C)                                                | 18587                             | 18215            | 18124            | 18051              | 17889              | 1.20                     | 1.21             | 1.21             | 1.21               | 1.21               |
| <b>64.4°F(18.0°C)</b>                                       | 18237                             | 17872            | <b>17783</b>     | 17711              | 17552              | 1.09                     | 1.10             | <b>1.10</b>      | 1.10               | 1.11               |
| <b>69.8°F(21.0°C)</b>                                       | 17449                             | 17100            | <b>17015</b>     | 16947              | 16794              | 0.85                     | 0.86             | <b>0.86</b>      | 0.86               | 0.87               |
| <b>75.2°F(24.0°C)</b>                                       | 16662                             | 16328            | <b>16247</b>     | 16182              | 16036              | 0.62                     | 0.62             | <b>0.62</b>      | 0.62               | 0.63               |

Bolded values were tested. All other values are simulated and for reference only.